

UNIVERSITY OF TWENTE

MASTER THESIS

Understanding social touch on
inflatable robots through gesture
recognition and personality
perception

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Abstract

This research explores how a round, inflatable soft social robot can recognize and respond to complex touch gestures, and examines how users perceive different robot personalities during interactions. By developing an inflatable prototype with a high-resolution touch-sensitive skirt using conductive threads embedded in silicone, this study investigates if it is possible to reliably detect touch gestures. Additionally, we examine if people are able to differentiate between personalities expressed by the robot. The gestures classified in this research include hugging, tickling, grabbing, squeezing, and stroking, and the studies personalities are defensive, calm, and excited.

Two studies were conducted. The first study collected and classified touch data from the robot’s touch-sensitive skirt, while the second evaluated user perceptions of three distinct robot personalities. In the first study, we show that a CNN-LSTM model can successfully identify gestures performed on the robot’s skirt. In the second study, we demonstrate in a between-participant study that people can differentiate between the robot’s three personalities based on Big-5 personality traits.

These findings establish a foundation for understanding embodied social touch in human-robot interaction and illustrate how robots can leverage their soft, deformable nature to create more intuitive and meaningful interactions. This work contributes to the HMI field by collecting gesture-touch data on a deformable inflatable robot and by assessing whether participants can distinguish between different expressed personalities.

0.1 Preface

I'd like to thank my supervisors for helping me through the turbulent times of my thesis and allowing me to explore and discover my own way. I wouldn't have been able to do this without them. Also massive thanks for my parents for supporting and believing in me, I love you. <3

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Chapter 1

Introduction

1.1 Background and Context

A social robot can be thought of as an autonomous robot that interacts and communicates with humans by following social behaviors and rules attached to its role, and with developments in Artificial Intelligence (AI) contributing to accelerating developments of their capabilities and applications, social robots are becoming increasingly visible in our everyday life. Existing robots like Neo ([1X Technologies, 2026](#)) will help with chores, the Tesla Bot ([Tesla, Inc., 2026](#)) will act as your waiter, and Vector ([ANKI, LLC, 2026](#)) will entertain your kids while you are gone. These robots are often constructed from rigid materials like plastic and metal, which shapes how they move and how people are expected to interact with them, and many of these social robots rely on anthropomorphic features, which can have several drawbacks. As a result, shape-shifting robots have been proposed as a solution to fill a void where humans and social robots are expected to grow closer and closer ([Schadenberg et al., 2025](#)).

Touch plays a central role in human social communication, especially in conveying affect, intent, and emotion ([Xu et al., 2022](#)), and so it should also be considered for social robots. With hard-shelled and anthropomorphic robots becoming the norm, physical touch is frequently treated as a secondary or constrained modality, rather than a primary means of emotional communication. For shape-shifting robots with soft or deformable bodies, touch is not only another sensory input, but an important aspect of their design. Despite this, relatively little work has gone into how touch-based interaction contributes to the perception of emotional communication in social robots, specifically within the context of shape-shifting robots ([Hedayati et al., 2022](#)). If we remove the recognizable smile, excited sounds, and known human and anthropomorphic features of the robot, are we still able to communicate and understand the emotions that a social robot is trying to convey?

This research builds upon prior work in social robotics by focusing on a soft social robot with minimal anthropomorphic features to explore the domain of touch and emotion. Through recent development in capacitive touch-sensitive technology for research and social robot purposes ([Englebienne and Sallaberry, 2025](#); [Teyssier et al., 2019](#)), this technology can be fitted to large, organic and stretchable surfaces. In our case, this technology can be applied to robots (See [Figure 1.1](#)), making the idea of soft social robotics like Baymax from Big Hero 6 an idea of reality.

To enable touch-based interaction, a social robot should be able to sense and recognize when it is engaged in a touch-interaction. Jung et al. (2017); Dutta and Kaboli (2024) already proved that it is possible to recognize different touch gestures, but much of the existing work still focuses on hard-shelled robots or uses limited ways of detecting and responding to gestures (Hughes et al., 2018; Müller Steffen and Gross and Horts-Micheal, 2018), so it is uncertain if gesture detecting works on a round, deformable soft robot. Additionally, we want to see if humans are then able to interpret the response during such a touch-interaction as well. By designing the robot with minimal anthropomorphic features we do not give any social cues through traditional anthropomorphic design or communication, which allows us to purely focus on the touch-interaction and a social-robot’s physical response to this touch.

In this paper we present an exploratory research based on a soft, round, non-mobile social robot with minimal anthropomorphic features, which we use as a platform to investigate whether touch gestures can be reliably distinguished on a soft and deformable surface. This information we can then use to provide an emotionally recognizable response to enable non-verbal emotional communication. While the robot may be able to recognize and differentiate between gestures during the touch-interaction, the response the robot provides might not be recognized in any recognizable way by the users. Therefore, the second part of this paper focuses on whether people can perceive emotional communication based on the response the robot gives during the interaction. We do this by giving the robot different emotional responses associated with different personalities through an expert design study and investigate if users can differentiate between these.

1.2 The robot

In this work we define a soft social robot as a robot that has minimal anthropomorphic characteristics, has no hard layer or components behind the touch-sensitive part as to allow for safe interaction with the robot, is pressure sensitive, is deformable due to the inflatable and stretchable nature of the skirt, is able to inflate or deflate, and shows social or recognizable behavior based on its movements and expressions.

Such a minimally expressive robot encourages the use of a users imagination and projection when interacting with the robot, and makes that it can only get emotional information from its movements rather than more common modalities. Using a minimal-anthropomorphic social robot stripped away from common emotional communication modalities that are often paired with anthropomorphic features has its benefits, like reduced expectation mismatch, reduced potential confusion or safety

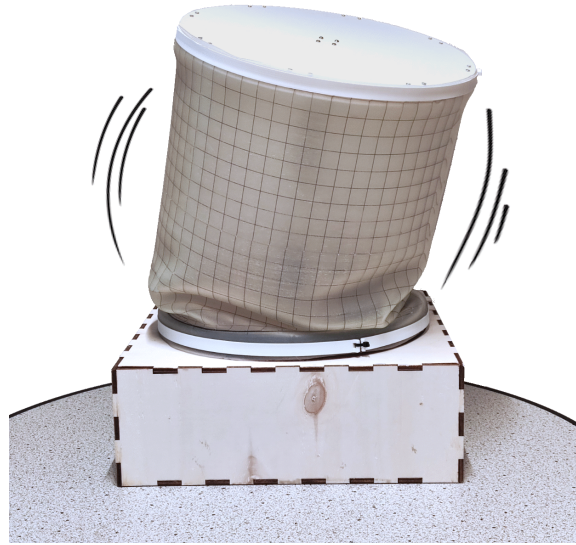


Figure 1.1: The physical appearance of the robot. Two pumps can inflate and deflate the robot.

risks, and reduced risks by reinforcing hu [et al., 2025](#)).

The robot used in this research specifically was developed at the University of Twente ([Figure 1.1](#)). The robot has 4 Degrees-Of-Freedom (DOF), and is physically controlled by three servos and two air-mattress pumps (See [Figure 1.2](#)). The touch-sensitive skirt is made from a high softness, highly stretchable Eurosil Shore A8 silicone, is ~4mm thick, and is fitted with a 27x19 grid-size capacitive sensor that spans the full 360 degrees of the robot. The robot has no front or back, no humanoid features, and minimal anthropomorphic features. The robot is placed on a wooden base that houses all electrical components, power supplies, and air-pumps.

To sense the touch we use the capacitive silicone technology presented by [Englebienne and Sallaberry \(2025\)](#). The capacitive points in the skirt have a gap of ~18 mm for a total area of 80x40 cm. This falls within the range of [Silvera-Tawil et al. \(2015\)](#)'s recommended 10~40 mm spacing and closely resembles the pixels per area found in the research from [Jung et al. \(2017\)](#); [Dutta and Kaboli \(2024\)](#) and [Müller Steffenand Gross and Horts-Micheal \(2018\)](#). Due to hardware limits we sample at ~7 Hz, which is slightly lower than the recommended 20 Hz by [Silvera-Tawil et al. \(2015\)](#).

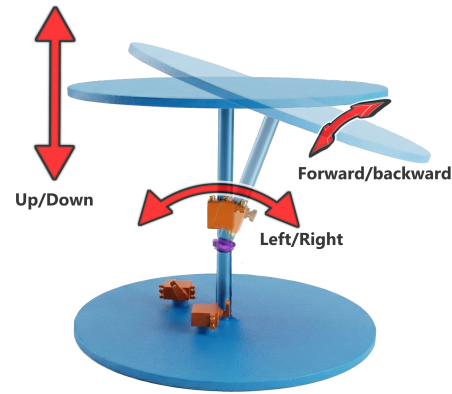


Figure 1.2: The movements of the robot. Three servos control the height, left/right, and front/back movement.

Chapter 2

Literature review

2.1 Behavior and personality of social robots.

To show emotional intelligence a robot should be able to both sense and display emotions. That is, the robot understands that it is being touched, and is able to respond to this touch ([Jung et al., 2017](#)). For this, we need to know how we can communicate different emotions from the robot to the person doing the interaction, and how we should design these personalities. The following related works provide the foundation for this.

2.1.1 Touch as a social interaction modality

Touch plays a central role in human social communication, especially in conveying affect, intent, and emotion ([Xu et al., 2022](#)). With rigid or anthropomorphic robots becoming the norm, physical touch is often treated as a secondary or constrained modality rather than a primary way to communicate emotions ([Hedayati et al., 2022](#); [Hughes et al., 2018](#)). [Schadenberg et al. \(2025\)](#) showed that (as an unwanted side effect) their inflatable social robot invited children to interact with it by touching an hugging. When this invitation to touch and interact with a soft social robot is intended, gestures like stroking, hugging, or tapping could be used as way of emotional communication.

2.1.2 Robot behavior

[Bretan et al. \(2015\)](#) claims that “it has already been shown that emotionally expressive animations have demonstrated effectiveness by making the agents more relatable and lifelike”. Making the robot act lifelike by adding animations are of essence when making social robots, especially if their intention is to work and communicate with real people. The gaze and postural shifts of a robot are equally important. Both can shift or encourage the flow of an interaction ([Sidner et al., 2003](#); [Cassell et al., 2001](#)). Even though our robot does not have a face, it can still focus attention by using its movement and posture. Characteristics like velocity, acceleration, and location also influence how people respond to the movements of the robot ([Laurel D. Riek et al., 2010](#); [Moon et al., 2013](#)).

[James J. Gibson \(1979\)](#) noted that observers usually assume some meaning of the visual appearance of a robot to its characteristics, like how size is associated

with speed (Anthropomorphism). Having a soft inflatable robot could evoke some of these assumptions as well, inviting people to touch it because it might look soft or inflatable. [John Harris and Ehud Sharlin \(2011\)](#) explored this idea even further by trying to convey emotion and intention by moving a stick in 1 DOF by varying its speed and direction. By using just these two characteristics, they were able to convey a wide variety of intentions.

2.1.3 Emotional communication through robots

[John Harris and Ehud Sharlin \(2011\)](#) designed a set of motions to test if robot motion alone (without familiar visual cues) can create strong emotional reactions in the participants. They found that organic motions, intended to be acyclic and more complex, tended to be recognized faster during energetic motions like angry and sad. These (e)motions include idle and happy, which could be matched to our own robot's personalities. [John Harris and Ehud Sharlin \(2011\)](#) research modeled idle motions with subtle pitching and rolling motions no greater than 10 degrees. For happy (energetic) emotions they used high-energy sequences while avoiding aggressive pitching motions.

Additionally, the research from [H .J. Eysenck \(1970\)](#) linked human extraversion and introversion to the activity-level they express. High activity mapped to extroverted characteristics, whilst low activity mapped to introversion. Therefore, a robot that expresses high activity through its movements might communicate emotions associated with an extroverted personality, whilst a robot that has a low activity level might communicate the opposite. This is confirmed by the findings from [Santa-maria and Nathan-Roberts \(2017\)](#), which found that extraversion-introversion is a dimension that is easily manipulated by researchers in the tested robot. Similarly, this dimension is also the dimension that user found easiest to report about.

Lastly, [Okuda et al. \(2022\)](#) also used three personalities to study the response of humans to a robot that responds to touch. Instead of creating a personality based on the extraversion-introversion domain, they tried to mimic human interactions. This approach is mainly based on (cultural) appropriate reactions to a hug, punch, or tickle, but because humans are very good at anthropomorphizing inanimate objects, it can still be very effective at communicating emotions.

2.1.4 Measuring social interaction

Self-reports are commonly used in social robotics to capture a participant's experience of interaction, and allow participants to express how they felt or interpreted these interactions. This is important because the success of a social robot is not only determined by how accurately it performs gestures or displays programmed behaviors, but by whether humans actually recognize and interpret these behaviors as intended ([Okuda et al., 2022](#); [John Harris and Ehud Sharlin, 2011](#)). And while the robot's responses are designed to communicate specific emotional states, the human participant may perceive these differently. Therefore, measuring personality rather than emotions allows us to see if it is possible to communicate emotions through a featureless soft social robot.

[Brandon \(2012\)](#) did a study on robot personality, where they used the BFI-44 to capture the perceived personality of their robot. This test has been used

in several other human-robot interaction studies, and has been shown to provide accurate and reliable results (Santamaria and Nathan-Roberts, 2017). Choosing the Big-Five test also provides consistency and interpretability to the results of the study, and by using the BFI we can explore whether the designed robot responses are able to communicate emotions. The BFI framework specifies five personality traits: *openness*, *extraversion*, *conscientiousness*, *neuroticism*, and *agreeableness*.

Within the BFI framework, *extraversion* reflects the amount of emotional expressiveness, which shows in verbal and physically active behavior. Low extraversion therefore corresponds to reserved, steady, or independent behavior. *Neuroticism* reflects the emotional instability one has, where high neuroticism corresponds to unpredictable or temperamental responses, and low neuroticism corresponds to stable emotional control. *Openness* refers to the relative depth of one's interests, and *agreeableness* refers to attributes like trust, kindness, and affection. Lastly, *conscientiousness* refers to the focus and level of self-discipline and self-control one exhibits (Howard, 1995; Cherry, 2019).

2.2 Touch-sensitive social robotics

Previous research about the application of touch-sensitive patches in robotics either mainly focused on the development of said patches (Teyssier et al., 2019), or focused on the application of (pre-made) touch-sensitive pads to already-existing (social) robots (Müller Steffen and Gross and Horts-Micheal, 2018; Hughes et al., 2018). A flexible silicone pad that can measure touch has been available for some time now, with early PCB prototypes showing up as early as 2008 (Park and Kim, 2008), and approaches that conform to robotic applications have been developed in the years after.

The core principles for touch-sensitive pads is the implementation of a conductive grid within a flexible silicone pad that creates a capacitive field. We can learn from previous research using this technology done by Teyssier et al. (2019); Hughes et al. (2018); Müller Steffen and Gross and Horts-Micheal (2018), and Englebienne and Sallaberry (2025) to create a touch-sensitive silicone skirt for our application. These all use Rx/Tx antennas to create a capacitive grid within a silicone patch to register touch. One of the advantages of having a capacitive field is the ability to detect field-disturbing objects (like a hand, skin, metal pole etc.) before it physically touches the silicone patch. Detecting non-conductive materials is only possible when they touch the sensor.

The research by Teyssier et al. (2019) is specifically interesting as they developed an organic touch-sensitive silicone skirt that is both easy and cheap to produce. By embedding conductive wire in silicone they created a patch that can detect touch on the pad. This touch-sensitive patch can be used for gesture and collision detection when applied to objects and robots, and preliminary tests have shown promising results using this technique to create patches for robots (Englebienne and Sallaberry, 2025).

2.2.1 Sensing techniques

Most touch-sensitive soft robotics in current research still heavily rely on external sensors like webcams, LIDARS, or (IR-)cameras to detect touch on a deformable

material. These technologies are expensive and require a lot of physical space, and are therefore not suitable for use as a touch-sensitive skirt on a (relatively small) social robot. For example, [Kwon et al. \(2012\)](#) uses a webcam and a software algorithm to calibrate and measure touch on a non-planar surface, [Andrew Stevenson et al. \(2013\)](#) uses an infrared camera, [Vogt et al. \(2004\)](#) uses a webcam with tracking points, and [Mackay \(2013\)](#) uses a depth-sensing camera.

[Jung et al. \(2017\)](#) reviewed 20 papers on social touch recognition, of which only two used capacitive pressure sensors in their design, both of which used a different and novel design to sense touch. [Troiano et al. \(2014\)](#) also reviewed 14 papers and their method to track gestures and touch over an elastic surface, and found that all of them either used an IR system, a simple camera, or a depth-sensing camera like a Kinect. The strength of these systems is their ability to work without a purpose-built (touch-) interface, their ability to provide multi-touch tracking, or their ability to do proper depth tracking instead of “pressure tracking”, but adversely do require a large amount of physical space, data and processing power to do so. As an example, the Khronos Projector ([Troiano et al., 2014](#)) used 1.100 tracking dots to compute the 3D touch map, which is both a major computational, technical and physical challenge.

Effectively detecting complex (multi-)touches and gesture patterns on organic and soft materials is therefore difficult. With not a single research from [Troiano et al. \(2014\)](#) using capacitive sensors, merely two studies from [Jung et al. \(2017\)](#) using capacitive pressure sensors, and only a few studies that used capacitive patches in hard-shelled robots, it seems that the use of a capacitive sensor patch inside a malleable inflatable surface is still very novel, and could be a solution for an otherwise complex and processing-intensive way to measure touch data on organic surfaces.

2.3 Gesture recognition

The gesture research done by [Troiano et al. \(2014\)](#) and [Jung et al. \(2017\)](#) both explored features for different gestures that were performed on deformable and capacitive sensor pads. The study done by [Müller Steffenand Gross and Horts-Micheal \(2018\)](#) is similar to this study, showing a hard-shelled socially assistive robot that was retrofitted with capacitive patches to sense when it is being touched or grabbed. It then uses a classification algorithm to identify the interaction (gesture) performed and responds. A systematic review done by [Tsirka et al. \(2025\)](#) analyses different gestures and touch classification algorithms used by similar studies, which shows that most of the classification algorithms used for gesture and touch-detection are based on supervised learning and recognize the touch in (almost) real time.

Many studies which provided gesture-recorded touch data were gathered through

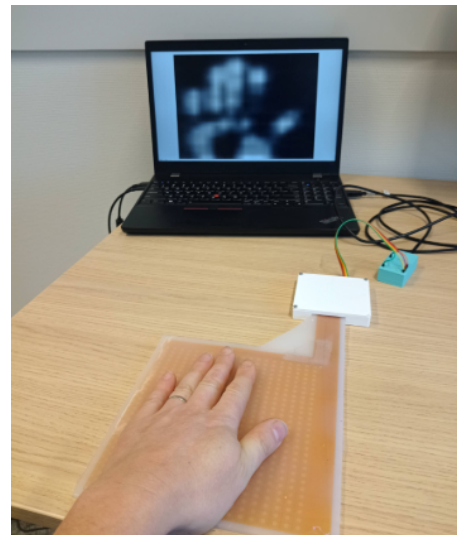


Figure 2.1: A silicone pad with resolution of 27x19 and its resulting Greyscale image ([Englebienne and Sallaberry, 2025](#))

rectangular touch-sensitive grids, usually placed on a hard surface and with a limited resolution or size (see Figure 2.2). For example, Jung et al. (2017) used an 8x8 point and 16x16cm touch-sensitive textile grid wrapped around the plastic arm of a mannequin to record their corpus of social touch (CoST). Dutta and Kaboli (2024) used a 9x9 point patch, and Müller Steffenand Gross and Horts-Micheal (2018) used a 16x16 grid to record gestures.

2.3.1 Gesture classification

Found studies used traditional machine learning methods to classify gestures. They are fast, reliable, and can work with the data directly from the touch grid. Jung et al. (2017) found that a Bayesian-, Decision tree-, SVM linear-, SVM RBF-, and the Neural network-classifiers all performed similarly, with an average ~6% difference per subject classification. The average accuracy was ~60% across 14 classifiers, which is impressive compared to the random guessing factor of ~7%. By neglecting to classify different intensities they were able to reach an accuracy of 78%.

The classification performed by Müller Steffenand Gross and Horts-Micheal (2018) (which was based upon the CoST by Jung et al. (2017)) also introduced a sliding window to capture time information per classifier. Implementing a sliding window adds extra information to the captured gesture and improves the classification, at the cost of some latency. Dutta and Kaboli (2024) showed similar use of classic classification models (k-nearest neighbor, random forest) to identify gestures. They also implemented a CNN and LSTM model, reaching accuracies of 97% over a guess-factor of 10%, outperforming the classic models. CNN's can classify touch independently from the learned locations, and LSTM's learn features that change over time (great for differentiating between a single and double tap). This combination of a CNN-LSTM model increases accuracy and speed when classifying.

However, these previously mentioned researches only tested gesture-recognition using capacitive touch-sensing on robots that were hard-shelled (Hughes et al., 2018), non-inflatable (Müller Steffenand Gross and Horts-Micheal, 2018), only allowed for low-resolution touch detection through simple on/off style pressure and touch sensors (Pandey and Gelin, 2018), focused on gesture recognition without the use of a soft social robot (Dutta and Kaboli, 2024), or did so using a square instead of a round patch (Jung et al., 2017).

The nature of our robot's soft, deformable and shape-shifting form introduces a variety of challenges. Studies on touch-based gesture recognition, such as those by Jung et al. (2017); Dutta and Kaboli (2024), typically rely on a hard backing with a fixed orientation and a clear reference. In contrast, the touch-sensitive skirt used in this work forms a continuous cylindrical surface without a directional reference, allowing gestures

to be continuously performed horizontally, which adds an extra level of difficulty when training a classification model. Furthermore, because there is no hard backing the soft skirt folds and deforms during interaction, resulting in interference within

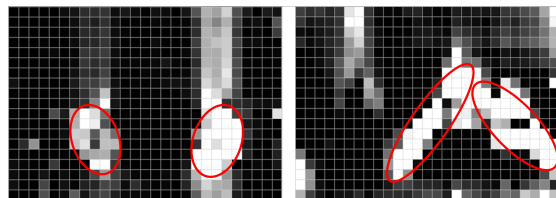


Figure 2.2: A 2-handed grab (left) and hug (right), introducing noise by deforming the skirt. Red circles are indicative of actual touch.

the capacitive skirt and introducing noise into the data (see [Figure 2.2](#)). Finally, the circular nature of our robot introduces a seam along the edge, which may cause gestures to appear split. These characteristics highlight a gap in existing literature, as current gesture-recognition approaches do not account for the combined effects of softness, deformation, and circular nature of the robot.

2.4 Problem Statement

The lack of touch recognition on social robots with minimal anthropomorphic features is a void we try to fill with this research. That is why we try to answer the question:

- How well can a LSTM-based classification model classify different touch gestures collected from a soft, round, deformable silicone skirt?

Additionally, we found little research or studies that explored the emotional response to these gestures. By providing our robot with different emotional responses based on different personalities, we can find if people can have recognizable emotional communication with a soft social robot with minimal anthropomorphic features. That is why the second questions we try to answer is:

- To what extent can people perceive different emotional responses from a social robot with minimal anthropomorphic features.

Chapter 3

Robot Design

3.1 Touch-sensitive skirt

The flexible silicone robot skirt went through 3 versions before reaching a final design. Each version used some form of a laser-cut peg-board covered with fork-shaped stand-offs. These standoffs were 3D-printed and were necessary to keep the wires spaced and distanced when the silicone was poured. The Rx and Tx wires were placed in horizontal/vertical lines, with each standoff being a crossing of the Rx/Tx wires. To cover more distance, the vertical lines made a single loop up and down, which increased the sensing area. The wires inside the skirt were the limiting factor when it comes to stretchability, as they stretch a maximum of 20-30%, whilst most silicone stretches a lot further (up to 500%). Therefore, the silicone used for the prototype was chosen because of its convenience and availability. The information of the touch-sensitive grid is extracted through an MTCH6303 with the use of an ESP32 micro-controller, which streams the data over a serial connection to a custom Python script. The script generates a live grey scale image of the touch-sensitive grid (See [Figure 2.2](#)).

3.1.1 Silicone skirt versions

Version 1

Version 1 used a round 3D cone-shaped mold and Eurosil Shore A4 silicone, and was meant to rotate and push the silicone against the wall. It failed as gravity pulled the silicone down during rotation, causing the silicone to snowball inside the round mold, which eventually failed this version. With this in mind, we adapted the future version to use a flat 2D design instead of a 3D design. (See [Figure 3.1a](#))

Version 2

Version 2 (and 3) were based on a flattened cone shape, produced by making a 2D mold and gluing the ends together. This version used Dragon Skin silicone, but as it is very viscous (thick, with little tendency to flow,) it resulted in some areas of the skirt being >6mm in thickness due to unequal spreading. This made the skirt heavy and rigid, which resulted in minimal movement and minimal inflation when pressurized. A new version that was thinner and more flexible was needed to solve these issues. (See [Figure 3.1b](#))

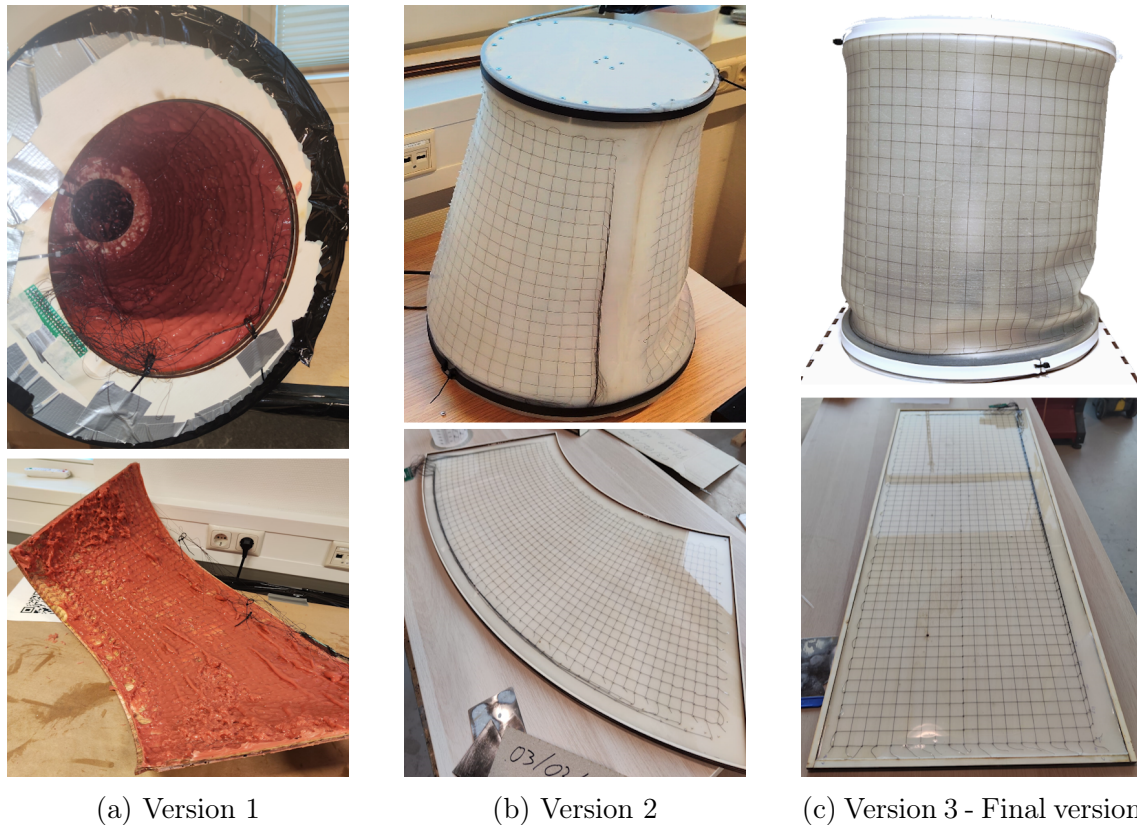


Figure 3.1: Different versions of the silicone skirt

Version 3

Version 3 followed the same production process as version 2, but by choosing a circular/rectangular shape the production complexity and time was greatly reduced. Eurosil Shore A8, being a less viscous silicone, was used to pour the mold, which resulted in a thinner $<4\text{mm}$ thick skirt all around. This version was lighter, more flexible, and more easily inflatable than the previous version, and was used during the studies. (See [Figure 3.1c](#))

3.2 Physical movement

The robot was placed on a base that housed the external electronics like the power supply, microcontroller, and inflating/deflating air-pump. For movement, we used three GXservo QY3240MG 40KG servos, powering the left/right, front/back, and up/down movement of the bot. For the fan, we used two generic 12v pumps commonly used to inflate air mattresses, as they strike a good balance between volume and pressure. These were all powered by a 5v power supply.

Chapter 4

Experiments

4.1 Part 1 - Recognizing social touch

The objective of Study 1 is to collect gesture data by having participants perform a selected set of gestures, which are recorded via the touch-sensitive skirt. Participants will be asked to perform gestures varying from a simple touch with their finger and hand to more complex gestures. The data that is collected during these interactions is then used to train a classification algorithm to identify the different gestures. Because the data from our inflatable skirt is low resolution, low frequency, and relatively abstract, no datasets were found that included gestures on a robot within this context. Therefore, we decided to collect our own dataset by having participants perform gestures on our robot and recording the sensed output of the skirt.

Based on the social-touch dictionary presented by [Yohanan and MacLean \(2012\)](#) we identified 7 gestures that 1) benefit from being performed on a soft inflatable and shape-shifting robot, and 2) have a distinct spatial and time-distributed pattern, which makes recognizing the different gestures easier. The resulting list of gestures are *hitting*, *hugging*, *tickling*, *grabbing*, *squeezing*, *stroking*, and *tapping*. Because the data from the inflatable skirt is low resolution, low frequency, and relatively abstract, we only need a limited number of participants (a minimum of at least 5 participants). Data augmentation can increase the amount and diversity of the data as well (see [subsection 4.1.2](#)).

[Jung et al. \(2017\)](#) and [Tsirka et al. \(2025\)](#) found no gender differences based on basic touch gesture characteristics. Age should be at least 18+ for safety, privacy, and consent reasons. Prior experience with robots is not a must. [Tsirka et al. \(2025\)](#) found no cultural correlation in their data, so cultural background is not considered. Therefore, the study will use a convenience sample, but with the aim of recruiting a diverse set of participants. For a step-by-step procedure and other considerations please see the full study design in [section A.1](#).

4.1.1 Data collection results

We collected touch-data over six participants (age 19-78, male and female), which resulted in a dataset where each gesture was included ~60 times, giving us a total of ~600 samples. This dataset can be found in [section A.3](#). The touch data was processed following the basic machine learning classification life cycle ([Krichen et al., 2022](#)), where we added a low-pass filter to remove noise. The frames were labeled

with one of the eight labels (The seven gestures plus none) to each frame as soon as there was direct contact with the skirt. Any frames outside of the touch were set as none.

4.1.2 Data augmentation and classification

To increase the number of entries we augment our dataset by shifting the recorded gestures. [Dutta and Kaboli \(2024\)](#) showed an improvement of $\sim 11\%$ when using augmented data in their model. Using a custom script we can set the number of augmented entries we want added to our original dataset. Possible augmentations include a wrapped horizontal, vertical and rotational shift of the recorded gesture, a probability of mirroring the whole entry, increasing or decreasing the speed of the recording, and increasing or decreasing the scaling of the recorded gesture. All modifiers use a random value between a set Min. and Max. value to generate the augmentations.

4.1.3 Model training

The research from [Dutta and Kaboli \(2024\)](#) used a capacitive 9x9 textile pad sensor, and trained a CNN-LSTM model to perform classification over 10 different finger gestures. They sampled touch finger gesture data from their 9x9 sensor at 15 Hz, having a higher pixel density, higher sampling frequency, similar resulting grayscale image, and similar flexible, although textile, material as our setup. They trained both a CNN and CNN-LSTM model, the latter performing slightly better on time-sensitive gestures (like single and double-tap). Their model was trained with 3000+ samples, achieving $\sim 97\%$ accuracy.

The combination of using a CNN, which allows for spatial feature extractions, and an LSTM, which adds the temporal time sequence processing, allows for the CNN-LSTM model to classify more complex gestures than most other ML algorithms. To achieve the best results, we chose to train a CNN-LSTM model. Training the LSTM-CNN model we followed basic classification training best practices. Labels and frames were split and separated over test, training, and validation sets. The data was split on a file-level to prevent skewed accuracy results. The models were trained using accuracy as its validation metric.

Based on the research of [Dutta and Kaboli \(2024\)](#), our CNN-LSTM model will use two time-distributed CNN layers with ReLu as an activation function, and a (2,2) Max-Pooling layer to reduce the size. Then we add a layer to flatten the data into a vector, which prepares it for the LSTM layer. The LSTM layer has 32 hidden units. A densing layer was added with the number of classes for our output. ([Shiwa et al., 2008](#)) suggests that for users to perceive a system response as directly related to their action, the delay should ideally be less than two seconds. A window size of 5 was used, which at 7 Hz sampling rate would mean a little less than a second of data. Window-sizes 7 (1 sec) and 14 (2 sec) were also checked to see if that would improve the model. Additionally, we experiment with the kernel filter and size, number of epochs, and the majority class reduction to find the best classification results.

The circular nature of the robot creates a seam on the sides, which can make a gesture appear split. This is not so much an issue when the dataset is large, but it is with our limited number of samples. By padding the image with the edges

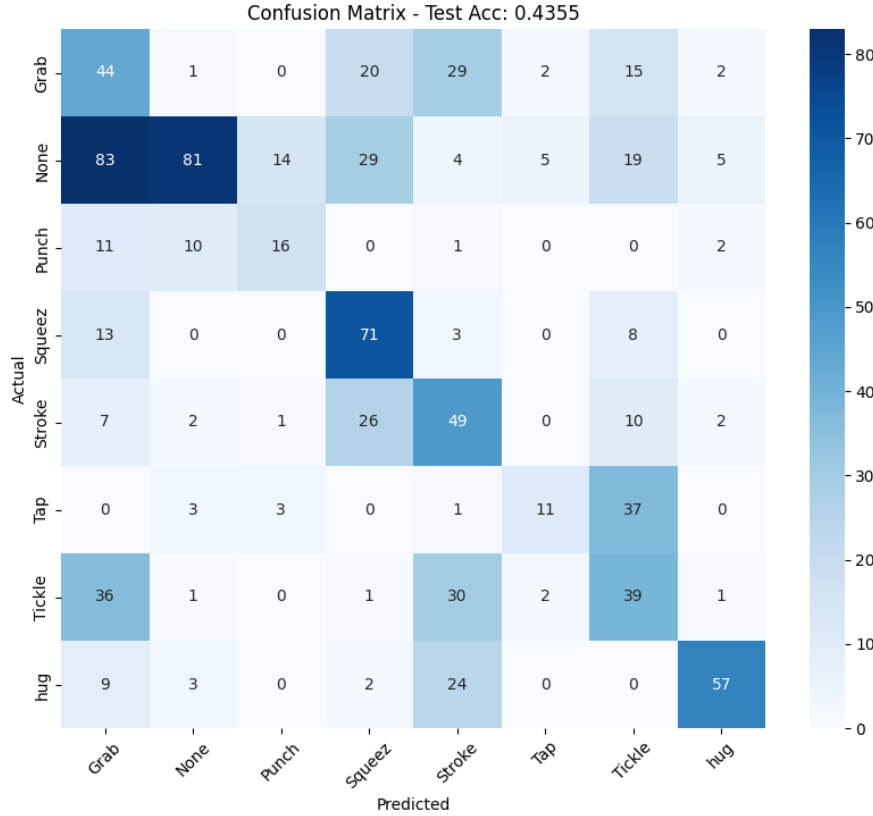


Figure 4.1: The LSTM confusion matrix results

from either side of the image, gestures can be artificially made continuous across this seam. However, especially around the edge, the image is quite noisy, as all the Rx/Tx antennas converge there. When padding the edges, this noise is also copied, which does not improve classification results. This was also confirmed in early tests, where padded models performed almost equally compared to models without padding. Therefore, padding was not used in our final implementation.

4.1.4 Training Results

The CNN-LSTM model was tested with different window sizes, strides, kernel filters and sizes to find the best performing model. When running a 5 frame window, stride 3, we found that the best CNN settings were a 32 5x5 and 64 5x5 kernel. In [Figure 4.2](#) we can see that the learning rate is plateauing after ~4 epochs, so we restricted the model training to 5 epochs. Class weights were added to prevent majority classes to skew the training results. This gave us a ~32% accuracy over a 28% guess-factor. When training the model with 15 augmentations per file (approx. 9000 samples) with a max random horizontal and vertical shift of 50%, we reached an accuracy of ~44%. When using a 7 frame window we reached a 46% accuracy, and with a 14 frame we achieved a 45% accuracy. The 5 second window therefore performed best considering its accuracy/frame ratio. When looking at the [Figure 4.1](#) we can see that gestures like a hug, which have very distinct time and spatial features, are recognized fairly accurate, while gestures like a tickle or grab have quite a large

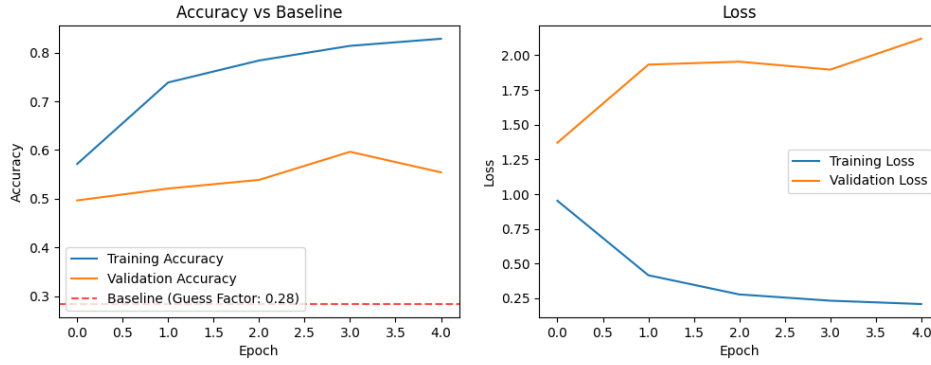


Figure 4.2: The LSTM training history

error. A grab gets mostly confused with the none class, which could be explained by the noise introduced during and after a grab, which is similar to the noise when no gesture is performed. The tickle gets mostly confused with taps, which makes sense as a tickle is just many taps. Some of the gestures are confused with seemingly unrelated other gestures like punch with none, stroke with tickle or a hug, or grab with a tickle. This could be due to small errors in the labeling of the gestures, where a frame might be labeled as none when the grab already started, or as grab while the gesture had already ended. In the end we were not able to fully pinpoint how or why these errors are present in our results.

4.1.5 Conclusion

We were not able to produce a well-enough working model, within a reasonable amount of time, that could classify, although marginally, gestures well enough during live interaction with the robot. With an accuracy of 44% our models have proven that we can classify gestures on a soft inflatable social robot, but this 44% accuracy has proven to not be good enough to be usable during live classification. During live classification trials the model was too slow and misjudged too often. Either extra data, a different approach, or more time is needed to create a working classification for the robot. Because of this, study 2 will use a wizard-of-oz- technique.

4.2 Part 2 - Measuring perceived robot emotions

Humans and robots interact with each other through a Human-Robot iteration loop (See [Figure 4.3](#)). The human-robot interaction during our study follows the same principle, where a participant expresses an emotion through a gesture (eg. slap, tickle, hug), the robot recognizing this gesture, responding with its movements to this emotion, and the participant interpreting this response. Therefore, for our second study, we want to find out to what extent people can perceive emotional responses from a social robot with minimal anthropomorphic features.

Emotions are interpreted through emotional communication, such as movement and posture ([John Harris and Ehud Sharlin, 2011](#)), and are often perceived as certain behaviours or as indicators of personality. Personalities are easier to measure than emotions because a personality test allows for neutral or no emotions being recognized, is a modality that is easy for people to self-report, and does not force

a decision of what emotions were seen. As our robot only has its 4-DOF to communicate emotion, we provide our robot with different movement patterns based on different BFI-44 personality traits. This allows us to use the BFI-44 to interpret the personality and the effectiveness of the robot to communicate or elicit an emotional response. The robot is provided with three different movement patterns, which can be found in [subsection 4.2.3](#).

We performed a between-subject user-study where each participant was showed only one of three personalities. We collected between-subject data from over 21 participants (MF, age 18-84, M=32, SD=19), 7 for each personality, from a varied pool of students, teens, seniors, elderly, whom mostly had no experience interacting with robots. During the experiment the participants were prompted to perform every gesture at least once, and were stimulated to interact with the robot until they had a good idea about its personality. After the interaction was over, the participants fill in a BFI-44 personality questionnaire about the different personality traits of the robot. If we can see a difference in the personality trait scores we can say that the participants have seen a difference between the personalities. For a step-by-step procedure and other considerations please see the full study design in [section A.1](#).

4.2.1 Measuring emotional communication

We will use a 5-point-Likert scale as is advised with Big-Five questionnaires. A dutch validated version of the BFI-44 will be used for participants who prefer dutch ([Denissen et al., 2008](#)). Questions will be asked as “I see the robot as a robot that...” to apply the questions to the perceived personality of the robot ([Brandon, 2012](#)). For example, “The robot is talkative” might result in a negative response from the participant, as our robot does not talk. However, asking the question as “I see the robot as a robot that is talkative” might produce a more fitting mental image about the robots personality, as the participant might find the robot outgoing and therefore it could be talkative.

4.2.2 Interview with puppeteering expert

Before we started designing our movements, we interviewed an expert in the field of puppeteering with robots in a research and theater setting. He recommended that, for a robot to have a personality, it should show this not only during the interaction, but also before and after. To achieve this we should think about implementing an (almost) continuous behavior for each specific responses/interaction/personality combination. An example would be to follow the direction of touch after or during the interaction. [Davo Sainz \(2021\)](#) did this by animating base behaviours, and blending it with the target animations during an interaction.

Additionally, he warned about the use of elements like breathing, which could lose its “magic” when being implemented as a basic movement. He emphasized that, when something as simple as a breathing pattern is used in a repeating and continuous cycle, it can quickly become dull and mechanical. To still allow for the robot to express something using inflation, you could incorporate it only during an interaction and make it depend on the personality (eg. inflate when hugging, in/deflate quickly when excited etc.).

For animating the robot, our expert suggested the use of puppeteering, allowing

for quick iterative cycles where the animation plane can be easily explored before and during the design phase. By adding smoothing and scaling the desired effect can be easily achieved after the recording is done, which is in line with the findings from [Davo Sainz \(2021\)](#).

4.2.3 Design of movement patterns

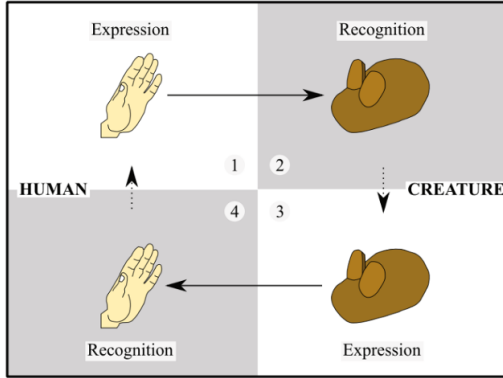


Figure 4.3: The Human-Robot Iteration (HRI) loop ([Cang, 2016](#))

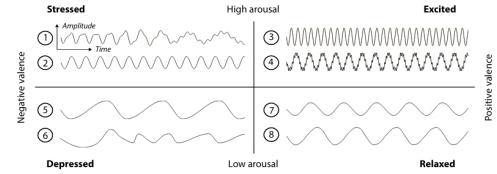


Figure 4.4: Breathing patterns on the arousal-valence scale [Bucci et al. \(2017\)](#)

We can inflate or deflate our robot through the use of airpumps. As breathing is a natural expression of emotion, one way to use this functionality is to imitate a breathing pattern and therefore help express these emotions ([Bloch et al., 1991](#)). There are three relevant respiratory aspects when it comes to modeling breathing: volume, timing, and irregularity in the breathing patterns. [Bucci et al. \(2017\)](#) did a user study which resulted in breathing patterns for our relevant personalities. These patterns can be seen in [Figure 4.4](#) and might be used during the design of the robot.

[John Harris and Ehud Sharlin \(2011\)](#) was able to convey a happy, idle, and angry personality and showed that movement energy and activity levels are commonly used to convey emotional within social robots, and [H .J. Eysenck \(1970\)](#) successfully linked the activity level to the extraversion, which is also easiest for users to self-report ([Santamaria and Nathan-Roberts, 2017](#)). Similar to the research of [Okuda et al. \(2022\)](#) and [John Harris and Ehud Sharlin \(2011\)](#), our robot will be given three distinct movement patterns directly linked with the BFI personality traits. One to communicate positive emotions, one to communicate neutral emotions, and one to communicate negative emotions. .

While the activity level of our robot determines how *extravert* our robot is perceived, *neuroticism* is associated with emotional instability, unpredictability, and negative affect ([Howard, 1995](#)). To be able to define our three personalities, we therefore include the *extraversion* and the *neuroticism* trait in our pattern design. By using all BFI questions we can identify if the designed movement patterns also have unique characteristics belonging to other traits. In this section we discuss how each movement pattern relates to the Big-Five personality traits. Each response style is described in terms of personality traits and movement patterns and can be found below. These are summarized as personalities of the robot and will be used in the rest of this paper.

Defensive personality

To communicate a negative emotion we designed a defensive behavioral movement pattern, which we refer to as the defensive personality. This personality shows avoiding behavior, recoiling as if repelled by the interaction, similar to how people instinctively pull away when tickled or touched inappropriately. In personality trait terms, this would be designed as high *neuroticism* and low *extraversion*. The robot actively moves away from the performed gesture while simultaneously deflating, and make threatening pitching movements to express its discomfort (John Harris and Ehud Sharlin, 2011). Movements are touch-aversive and fast.

Neutral personality

To communicate a neutral, uninterested emotion we designed a neutral behavioral movement pattern, further mentioned to as a neutral personality. This personality detects a touch or gesture but moves minimally, acknowledging the interaction without strongly responding. The goal is to create a reaction that feels like the robot recognized the touch, yet is apathetic in its response to the interaction. Based on the personality trait this means we aim for medium *extraversion* and medium *neuroticism*. This translates to minimal movements when touched, maintaining a stable emotional level, and balancing between defensive and proactive behaviours (John Harris and Ehud Sharlin, 2011). Movements are slow and subdued.

Excited personality

To communicate a happy emotion we designed an excited behavioral movement pattern, further mentioned to as the excited personality. It depicts a friendly, excited robot, and depending on the type of gesture, the robot will physically lean into the gesture and show excitement during or after being touched. In terms of Big-Five personality traits, this translates to high *extraversion* and high *neuroticism*. The robot moves or leans toward the source of the gesture, and the movements are fast, energetic, and exaggerated (John Harris and Ehud Sharlin, 2011).



Figure 4.5: Puppeteer setup: Robot (left), Computer (middle), Omega7 (right)

4.2.4 Designing gesture responses - An expert study

To design the actual movements for each of the movement pattern we met up with three experts in the field of shape-changing social robots, social robotics, and theater.

We used an 4-DOF Omega 7 ([Force Dimension, 2025](#)) that was programmed to puppeteer the robot. With this setup we could reenact, discuss and iterate on the responses until we were happy with the result. A picture of the setup can be seen in [Figure 4.5](#). During a two-hour session we discussed and designed responses to all of the seven gestures for the three personalities, as well as three base behaviours when the robot was in rest. Each of these was (if applicable) designed for both left- and right-side touch. The result of this session were 34 responses to touch (including left/right variations) and three base behaviours, which can be found in [section A.3](#). We validated the movements of each personality with two of our design experts. A list of our observations during the design process can be found in [section A.2](#).

During our literature review, we identified breathing patterns that could suit our robot’s personality. However, after careful discussions with our experts, we decided not to include them in the design. Instead, we found that inflating and deflating was better suited for expressing emotion rather than being reserved for breathing, and to avoid the robot appearing repetitive or overly robotic, we decided not to use breathing patterns at all.

4.2.5 Technical setup

We chose the Omega 7 as it was accessible to us, has 4-DOF, and had a workable python SDK available. We programmed an interface that took the Z-axis, Roll and Pitch values from the Omega 7, mapped them to the right scale, and forwarded these values to the robot. We could scale the range of movements, and map the values over time to be easily converted to a three channel input mapping for storing and executing. An example of a mapped response can be seen in [Figure 4.6](#).

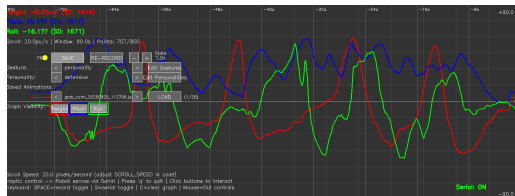


Figure 4.6: The movements of the robot

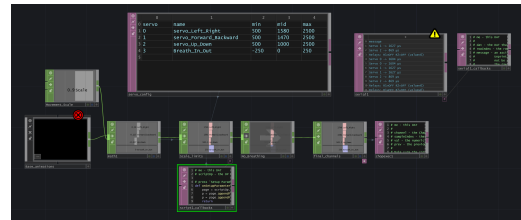


Figure 4.7: The physical appearance of the robot

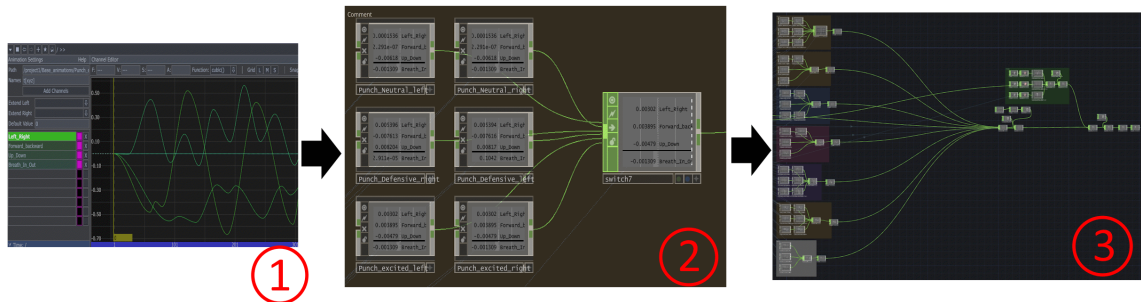


Figure 4.8: TD response and behavior design (See [section A.3](#) for all designs)

4.2.6 TouchDesigner implementation

To program the robot’s response movements, we used TouchDesigner (TD) (Derivative, 2025), drawing technical inspiration from Davo Sainz (2021). We took the gesture mappings generated during the puppeteer session (Figure 4.6) and converting each into an animation in TD (step 1 in Figure 4.8). These gesture animations were then combined with the personality variations, including their left/right counterparts, into grouped modules (step 2). Next, the groups for each gesture response were merged (step 3). All resulting groups were connected to a switch operator, which can be toggled using a keyboard or controller interface to trigger the desired response. Finally, we mapped the three base behaviours and combined them additively, following the blending approach described in Davo Sainz (2021). Both the base behavior and gesture response can be scaled accordingly ($\times 0.9$ and $\times 0.8$ in our case) to increase or reduce sensitivity to the mappings.

Next, the channels are pre-processed and converted, ready to be sent to the robot (See image Figure 4.7). The incoming channels have a range between 0 and 1, and are mapped to a PWM value between 500 and 2500 depending on the servo limits. The limits and center points are set in a CSV-file, and a custom script (??) takes the incoming values and maps them to the proper PWM values. These values are then sent to the robot.

4.2.7 Wizard-of-Oz

Study 2 was performed using a Wizard-of-Oz technique. The robot was set-up around shoulder-level for each participant (see figure Figure 4.9). During the interaction between the participant with the robot, the researcher trigger the pre-recorded animation to respond to the touch. The researcher was positioned away from the setup to suggest that they were interested in the laptop/program. Multiple participants mentioned that they were not aware that the robot was being controlled by the researcher, suggesting that the participants assumed the robot moved autonomously. After the participant has seen all the different responses to the gestures, and after they indicated that they had seen enough of the robot during the interaction to get a good idea of the bots personality, we had them fill out the BFI-44 questionnaire.

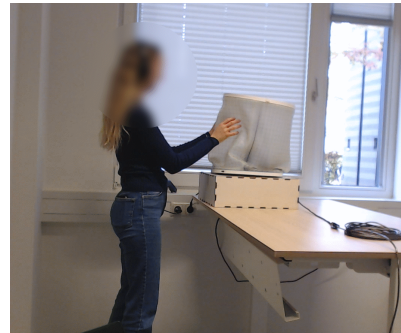


Figure 4.9: The setup for study 2

4.2.8 Modeling approach

To answer our research question, we take the mean score per personality trait of each participant, and create a box plot with the data (See Figure 5.1). To make claims about our results, we will have to estimate how our data behaves over a larger population. As we do not know our population shape in advance and every movement pattern produces a different distributions we assume a Beta distribution. We fit a univariate Bayesian model to our data, using a formula which models the relation between the personalities and the score in each trait using a Markov Chain Monte Carlo (MCMC). The model then makes posterior predictions based

on the personality and personality trait. Based on the earlier assumptions we can then infer the posterior draws, which represent the probability of our uncertainty about the expected score for each personality and trait. By computing the posterior distribution of the score differences, we can say how likely it is that personality A would score higher than personality B on trait X. These results can be found in the results in [Table 5.1](#).

We also fitted the data to a Multivariate Bayesian model to compare the five personality-trait responses across the whole robot personalities. This type of model estimates the correlations between the five traits for each personality (e.g., someone who saw the excited personality will usually score high in extraversion and openness). A Beta distribution was again assumed. The posterior draw generates a five-dimensional vector for each robot personality. Using the results from the posterior draw, we calculate the Euclidean distance between the vectors of two personalities (e.g., between the excited and the calm personality) to get a between-personality distance. We also take the average distance within the vector distribution of each personality to calculate the within-personality distance, which we use as a threshold. If the between-distance of two personalities is larger than the within-distance of the two personalities, we can say that they are different.

Chapter 5

Results

5.1 Perceived emotional recognition

The boxplot in [Figure 5.1](#) shows the average 95%-CI score the participants gave to each personality, based on each personality trait. The personalities are order sequentially in the *extraversion* trait, with the excited personality scoring highest, followed by the calm and defensive personality. Interestingly, the calm personality scores almost equally to the excited personality in the *agreeableness*, *extraversion*, and *openness* trait, even though the behavioral patterns differ significantly. This suggest that people need minimal emotional cues to interpret a robot as being extravert, agreeable, and open. We can also see that all three personalities score relatively equal in the *conscientiousness* trait ([Table 5.1](#) [Figure 5.1](#)), meaning not a single personality was perceived focused or as having a goal ([Howard, 1995](#)). We did not intend this with any of our personalities, so this is expected within this trait.

	Agreeableness	Conscientiousness	Extraversion	Neuroticism	Openness
Excited > Defensive	0.989	0.625	1.000	0.003	0.946
Calm > Defensive	0.996	0.745	0.853	0.528	0.780
Excited > Calm	0.351	0.376	0.993	0.002	0.803

Table 5.1: The probability that one personality scores higher within a certain personality trait when compared to another personality

When doing a quantitative analysis on these observations we can say something about how certain we are that the personalities are actually being perceived differently. [Table 5.1](#) shows the probability that a personality scores higher for a personality trait compared to another personality, and thus how certain we are that the personalities per trait are different. However, to see if users were able to recognize differences in the whole personality of the robot we compare the five personality-trait responses across the whole robot personalities. We do this by fitting the data to a Multivariate Bayesian model. The resulting probabilities are shown in [Table 5.2](#). We can see there is a high probability that our participants can tell a difference between excited and defensive, as well as between calm and defensive, and a slight certainty of ~69% probability that they can tell the difference between excited and calm.

Based on the certainty shown in [Table 5.2](#), we can say that the participants successfully interpreted and perceived emotional communication. The high proba-

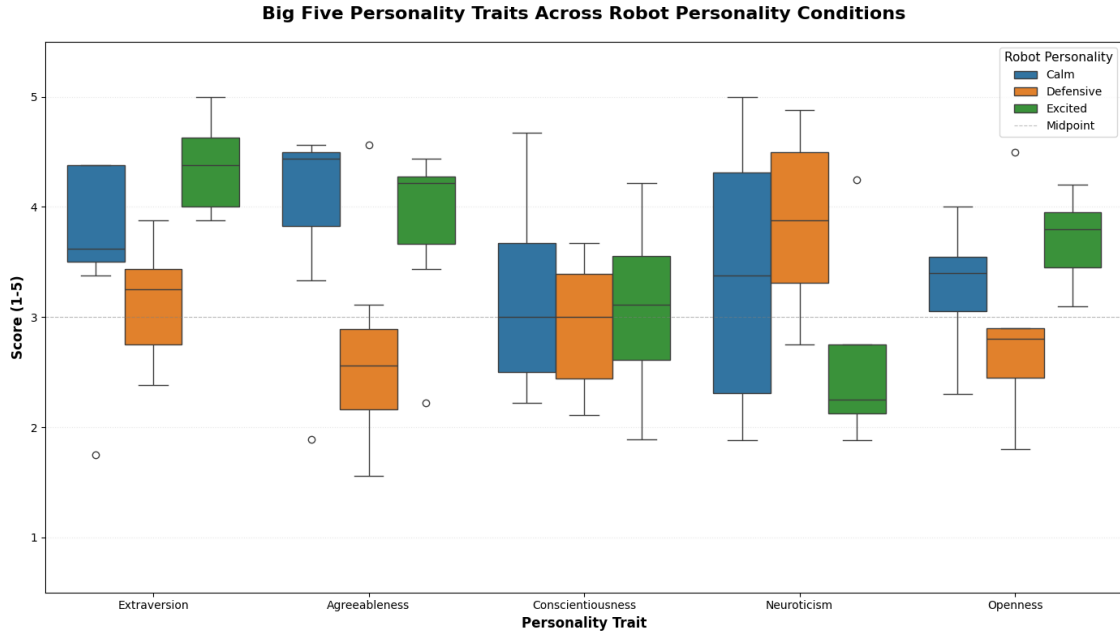


Figure 5.1: 95%-CI Average score per personality per BFI personality trait

bility that participants distinguished the defensive personality from both the calm and excited personalities suggests that the participants perceived the negative emotional response, and the lower probability between calm and excited shows that the participants did perceive a difference in the emotional response, although with less certainty. This means that movement alone is enough to communicate emotions.

	Probability
P(Excited - Calm distance > typical within distance)	0.693
P(Excited - Defensive distance > typical within distance)	0.969
P(Calm - Defensive distance > typical within distance)	0.974

Table 5.2: Probability that one personality is perceived more different then another personality

To see if participants could judge which personality they saw based on their interactions we asked them for their guess on the personality after they filled in the BFI questionnaire. This measure was only used for qualitative observations, and therefore we did not control for guessing. Some participants mentioned that they thought that the movements of the calm personality were fast and/or large, thus suggesting an excited personality to them. This can also be seen in their guesses (Figure 5.2, base frequency of 7). The defensive and excited personality were mostly guessed correctly, but the calm personality was more-often guessed as either excited or defensive then calm. This is also seen in their responses on the BFI-44 questionnaire. We also trained a Multinomial logistic regression model on the collected data. Based on the scores a participant got in each personality trait, this model made a prediction on the personality they saw (Figure 5.3). This shows a distribution that is more in line with the actual personality that was shown, confirming that using the BFI-44 is a more accurate way of measuring personalities compared to just giving the participants three personalities to choose from.

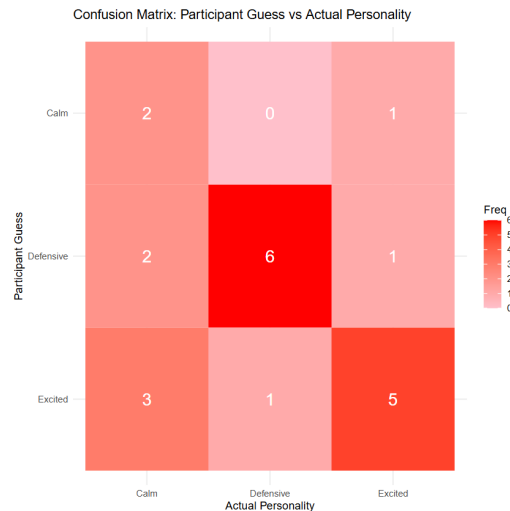


Figure 5.2: Comparison of participant guess and actual personality shown

Our findings in Figure 5.3 also show that participants who saw the calm personality often confused it with the defensive and excited personality. This is in line with the observations from the boxplot, where we can see that calm and excited score similarly high on *openness*, *agreeableness* and *extraversion*. Similarly, Table 5.1 also confirms that there is a low confidence in the *agreeableness* trait between calm and excited. Calm also has a large overlap with the *neuroticism* trait when compared with the defensive personality.

In Figure 5.4 we compare the self-reported personality against the predicted personality based on each participant's answer. This shows that users mostly identified the robot defensive both in their personal understanding as well as by their guess, often rated the robot as excited even though their answers suggested the calm personality, and they usually did not perceive the robot as very calm. This is in line with our previous findings when looking at the scores from the excited and calm personality traits in Table 5.1, Table 5.2, Figure 5.3, and in Figure 5.2. These all show that participants who saw the calm personality often guessed the excited personality and thus being hard for our participants to differentiate between.

5.1.1 Qualitative observations

General observations during both user studies were that we noticed that some people heavily projected their/a pet onto the robot. Especially since the robot was motionless during the first study did we see this. We also saw this phenomenon during the second study, however less extreme. This was especially true for participants that projected heavily during the first study and returned for the second study, where the robot now expressed a personality.

General observations during the second user study were that multiple participants mentioned that, for all three personalities, the robot looked somewhat "autistic". They mentioned this because the robot sometimes seemed unsure or slow to respond, recognizing autistic behavior in this. We also noticed that the users could control the robots behavior with their own; when being fast or withdrawn in their interactions, so did they observe the robot doing this. Because of the loose skirt of

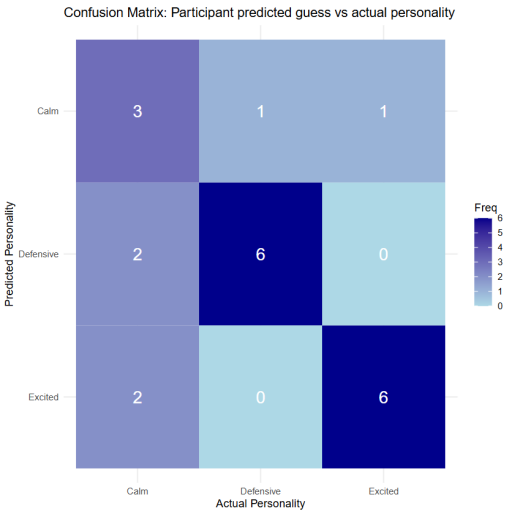


Figure 5.3: Comparison of BFI-44 responses and participant personality Guess

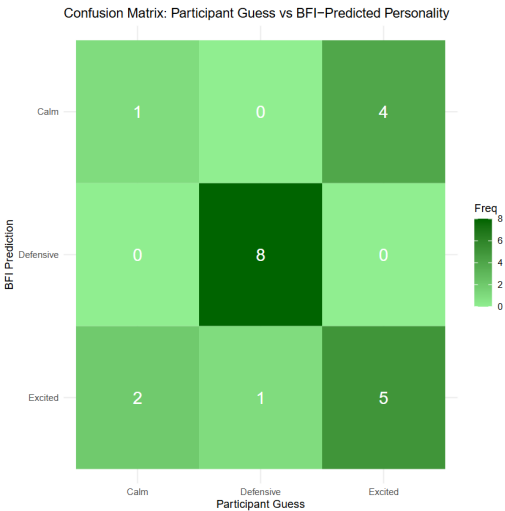


Figure 5.4: Comparison of BFI-44 personality prediction and participant guess

the robot, quickly touching it might make the robot also appear high in energy. This did cause that some participants to have guessed a different personality then they had seen. This is also reflected in the data ([Figure 5.2](#)).

5.2 Discussion

5.2.1 Perceived emotional communication

Interestingly, the results from [Figure 5.1](#) show that the trait’s scores for the defensive personality are clearly distinct from the rest, whereas the calm and excited personalities scored very similarly. This similarity is interesting because the movement patterns, responses, and base behavior of the calm and excited personalities were designed to be quite different. Both personalities received similar scores on the *extraversion*, *agreeableness*, and *openness* traits, even though the participants could not compare the personalities. Apparently our participants still thought of the robot as being quite energetic and active, even though the movements were considerably less energetic compared to the active personality. We explicitly designed the calm personality to communicate a neutral, uninterested emotion, which is also reflected in the high variance that can be seen in the neuroticism score.

An alternative explanation for the similarity between the calm and excited personality could be that, based on comments we received from our participants, the calm personality was not calm enough, and should respond even slower and less extreme to input during interactions than we had it programmed to do. This claim could also be supported by the research from [H. J. Eysenck \(1970\)](#), who linked human extroversion and introversion to the activity-level they express. Additionally, when participants quickly touched the calm personality robot, the robot quickly jerked with each touch, suggesting a higher energy response than what it was initially programmed to.

Also noteworthy is the *neuroticism* score of the excited personality. We aimed to design both the defensive and excited personality to have high neuroticism, but this is only reflected in the defensive personality. During the expert design process ([subsection 4.2.4](#)) we interpreted high neuroticism as having a resilient and temperamental response to touch; large and excessive movements when responding to touch. However, upon further investigation, neuroticism in the BFI-44 framework mostly adapts to the negative emotions present in someones personality ([Howard, 1995](#)). Participants who saw the excited personality did not assume any negative emotions, intentions, or over-reaction from the robot, thus scoring low in this trait. Alternatively, the defensive personality evoked largely negative emotions, thus scoring high on the neuroticism trait. With the calm response we aimed to create a balanced and neutral response, which is reflected in the large spread seen in the neuroticism trait. It reflects that participants were uncertain about its energy being positive or negative, which was our initial goal with the calm personality. In the end, all three personalities reflected the amount of “negativity” in their neuroticism scores.

5.2.2 Gesture classification

Even after spending a considerable amount of time to obtain a CNN-LSTM classification algorithm that could work in a live setting, we did not succeed in doing so. This was due to the fact that the data was collected on a stationary robot, and the live classification was done on a moving robot and skirt, which made it harder to get a good classification. We also collected too little data during the first study. For example, [Dutta and Kaboli \(2024\)](#) collected over 3000 samples before augmentation, whilst we only collected only ~600 samples. We expected that augmenting the data

would produce enough extra training data, but this did not improve our accuracy with any considerable amount. We tried many different approaches and settings, but we were unable to get a use-able model. Additionally, short gestures like tap, grab, and punch, should be over-sampled, as they produce only a marginal amount of data compared to longer gestures like hug, grab or tickles. We do think that, due to the noise and the low-resolution Grey-scale image the touch-sensor produces, we don't need more participants to achieve this. Simply collecting more touch data from the participants would suffice.

5.3 Conclusion

Our first objective in this research was to investigate whether we can detect different touch gestures on an inflatable soft and round touch-sensitive silicone skirt. By designing a 27x19 resolution capacitive silicone skirt around a movable, round soft social robot, we collected data for seven touch gestures from six participants. Using LSTM and CNN techniques, we trained a CNN-LSTM classification model that was able to recognize different touch gestures on a soft inflatable skirt with an accuracy of 40%. Because the model responded with low accuracy and high latency, it was not suitable for autonomous response triggering during the second study, which led us to use a Wizard-of-Oz approach for personality-based interactions. Even with these limitations, the results show that gesture detection on an inflatable, soft, and round robot surface is feasible.

Our second objective was to investigate whether people can differentiate between a soft inflatable featureless social robot's personalities when interacting with it. Through an expert design study, we created three personalities based on basic BFI-44 personality trait goals and translated these into motion responses for the robot, addressing how a soft social robot should react to different touch gestures. A between-participant study then showed that our participants were able to differentiate between the personalities both on individual traits (Table 5.1) and on an overall personality level (Table 5.2). Through this second study, we demonstrated that people are indeed able to recognize and distinguish personalities when interacting with a soft, featureless inflatable social robot.

Throughout this project, we developed a soft social robot that uses a capacitive touch-sensitive silicone skin, is capable of collecting touch and gesture data, can classify multiple gesture types, and can respond to users in recognizable and personality-driven ways. Together, these findings help fill the gap in touch recognition and personality expression for soft inflatable social robots.

With this research we showed that it is possible to communicate emotions that people can interpret and distinguish from each other through a soft robot with minimal anthropomorphic features. We show that it is not necessary for future social robots to comply to humanoid, zoomorphic or anthropomorphic designs or common communication modalities like speech, facial expressions or screens, and designs can move towards embodied behavior and physically active designs where touch and physical movements can provide a source of emotional communication. We showed that people are able to distinguish between negative and non-negative emotions through different movement patterns, which makes personalities based on different movements patterns a valid communication method.

We also showed that touch can be a medium that can enable this type of communication. Our participants recognized an emotional response from a soft social robot with minimal anthropomorphic features after having performed a physical gesture, which shows that using touch can be used for communication. Simultaneously, we showed that both these findings can be performed on a minimalist soft social robot, which could make these types of robots, touch modality and emotional response a great alternative to common humanoid robots.

5.4 Future work

There are a few key areas in this research which could be improved upon. Regarding the design of the robot, future work could improve on the stretchability of the conductive material within the silicone skirt. The copper threads prevented the skirt to inflate fully, so more stretchable material, like flex-PCB's, could improve this aspect. A more flexible skirt could make the robot less damage-prone, and gestures like punches and hugs could benefit from the added stretchiness. Similarly, more attention could be given to getting better, more silent high-flow high-pressure air-pumps to quickly and silently inflate the skirt, which could then be used in the behavioral movement design of the robot. The servos could be replaced by stepper motors to make the robot move smoother, quicker, with less noise, and with more power to manipulate the shape of the stretchable silicone skin with more control, whilst allowing participants to engage with the robot unrestricted.

Next, even though we achieved a classification accuracy that was higher than the guess-factor, we still believe that there is a lot of room for improvement. By using a higher resolution grid to sense touch it could increase how much data is collected during a touch, improving the classification. Similarly, collecting data at a higher frequency could result in much more data and lower latency when detecting touch. Due to the deformability of the skirt, when an interaction happens, a lot of noise in the data is introduced as the carefully calibrated capacitive fields now interfere with itself. By filtering or reducing the noise during data collection and live classification the accuracy could improve, possibly by using a different material as a substrate for the antenna wires. Combining this with better models should result in a classification model that could be used in a live setting.

Additionally, there is a possibility to integrate a camera or an air-pressure sensor to this setup to add extra dimensionality to the data, which would improve the gesture recognition model further. The movement of most gestures already start before the physical touch has even happened (like arms opening for a hug), and with the use of cameras on the robot these gestures could be detected better and earlier. Then, when the robot reaches a well enough working live recognition model, study 2 could be repeated without the use of the Wizard-of-Oz to see if the same results are being achieved when the robot responds autonomously instead of manually.

Lastly, future work could focus on the use of a soft inflatable social robot in a social context. During this research, we always required our participants to touch the robot. However, it would be interesting to see what behavior a soft social robot evokes when placed in a non-research and controlled environment, especially compared to a normal robot. It would be interesting to see if people would prefer a soft robot over the hard-shelled robots that come to market nowadays, or if they are able to build long(er)-term relationships with these types of robot (just like Baymax in *Big hero 6*). Similarly, future research on personality in human-robot interaction could design experiments that measure both human and robot personality when working with a soft inflatable social robot.

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Appendix A

A.1 Study notes

A.1.1 Study 1: Data Collection

Objective

The objective of Study 1 is to collect gesture data by performing a selected set of gestures that are both recorded through the touch-sensitive skirt and the camera. This data is then used to train a classification algorithm to identify the different gestures. Participants will be asked to perform gestures varying from a simple touch with their finger and hand to more complex gestures like slap, punch, pinch, hug, and tickle. For the full list of gestures, see [chapter 4](#).

Participants

Minimum of five participants. As the gestures are easily distinguishable from each other, little data is needed to train the classification. Data augmentation can increase data points as well. If early trials identify gestures previously not considered, we can increase the number of participants needed to gather this data. The data collection will mainly take place in (a private room in) the HMI lab.

Participant demographic

[Jung et al.](#) and [Tsirka et al.](#) found no gender differences based on basic touch gesture characteristics (SD of 0.01%). Age should be at least 18+ for safety and consent reasons. Prior experience with robots is not a must. [Tsirka et al.](#) found no cultural correlation in their data, so cultural background is not considered. Therefore, the study will use a convenience sample, but with the aim of recruiting a diverse set of participants.

Data collected

- Labelled sensor data from the capacitive skirt
- Labelled video data from the camera

- Participant feedback

Procedure

1. The participants sign the consent form and are given a brief explanation of the robot, its touch-sensitive skirt, and what data will be recorded.
2. Before any further instructions are given, the users are encouraged to engage with the robot to familiarise themselves, as well as:
 - They are asked to think of gestures they think they can perform on the robot. This provides us with a set of gestures we might not have thought about, and we can decide on the fly to include in our dataset.
3. Participants are shown examples of the gestures they will be asked to perform (e.g. slaps, squeezes, pinches, hugs, tickling, punch).
4. They are instructed to perform each gesture naturally as they would in a social interaction and with different intensities.
5. Each participant performs the touch gestures on the robot's surface.
6. The gestures are recorded in a sequence per gesture using a custom script. The data from the skirt (pressure, duration, location, intensity) and the camera system (visual recognition of the interaction) are recorded.
 - Each gesture is repeated multiple times to ensure robustness and more data during the data collection.
 - Each sequence is immediately annotated with the right label.
7. Participants provide feedback on their experience interacting with the robot.
 - The main focus of this part is to catch any thoughts the participants have on the setup, performance or gestures. No specific knowledge or questions are prepared.

Considerations

- Participants will perform gestures at varying intensities, enhancing the robustness of the trained classification algorithm. This variability ensures that each gesture classification accounts for different durations and intensity levels.
- Each data entry is recorded through a script and will have the label saved with it.

A.1.2 Study 2: Evaluating Robot Responses

Objective

The objective of study 2 is to see what robot behavior responses are suitable for what gesture. By asking the participants to perform a certain gesture, they will be shown one of three random behavioral responses from the robot: either a defensive, Calm, or Active response. These will either be triggered through automated detection, or using wizard-of-Oz technique. The participants fill in a short questionnaire after each interaction to map their thoughts about the interaction. To find an exact description of the patterns, see [chapter 4](#).

In total, there are 7 gestures x 3 behaviours, which will result in 21 interactions tested. The technical implementation will be based on TD and findings from [Davo Sainz](#), and refined during an expert design session ([subsection 4.2.4](#)). During the study we will show one personality per participant, who will interact and see all seven gestures and responses we designed for that specific behavior during our expert design study. We will use the BFI-44 ([Howard, 1995](#)) questionnaire to analyze the responses from the participants on the robot's behavior (see [subsection 4.2.1](#)).

Participants

Because this is a between-participant study taking only 15~20min, we aim at getting 21 participants. seven for each robot personality.

Demographic

The same requirements as Study 1 apply.

Data collected

- User BFI-44 ratings of the robot's responses (1-5 Likert-scale responses).
- Open-ended feedback on the robots personality.

Procedure

1. Participants sign the consent form and are given an overview of the study.
2. Before the official study is conducted, the participants are allowed and encouraged to engage with the robot to familiarise themselves.
3. Before starting, they are asked if, for any reason, they do not want to perform specific gestures due to personal beliefs or discomfort with certain gestures.
4. They are asked to interact with the robot, performing each gesture/interaction.

5. The robot responds based on its the performed gesture and pre-programmed response.
6. After being shown the behavior of the robot, participants fill out a the BFI-44 questionnaire, as well as provide a guess on the personality the robot displayed (eg. calm, defensive, excited).
7. After the study is finished, participants will provide answers to a short, structured interview to gather qualitative insights about the personality of the robot. These questions are not formally noted, but if answers surprise us we will note them down ([section 5.2](#)).
 - “What did you think about the personality of the robot?”
 - “Where there any responses that you found (not) fitting?”
 - “Which response felt the most natural, and why?”
 - “Which response was the most unexpected, and why?”
 - “Would, and if so, wow would you expect a different response, and why?”

Considerations

- If study 1 results in an improper classification, it is possible to trigger the responses in a wizard-of-oz style, as the performed gesture is known beforehand.
- If participants systematically choose “defensive” as engaging but “neutral” as most appropriate, you can separate entertainment bias from actual gesture-response fit through the use of the questionnaire.
- A video will be recorded during the study.
- The stored video can also be re-watched when ambiguity arises on the collected data or when extra context is needed.
- Participants are fully briefed about the goal of the research (collecting touch data/identifying the best response to advance social robot technology), both in advance when recruiting, as well as before, during and after the study itself. The purpose of the research is not difficult to understand for most of the audience, but for people who are unfamiliar with robots or HMI, extra time will be used to make sure they fully understand the goal and tasks. Participants are also free to ask questions about the research and the goal before, during and after the studies. A consent form with information about the gathering and processing of their data is provided beforehand, and an explicit statement is made verbally and on the consent form that the participants are free to withdraw at any moment without explanation.

A.2 Expert design observations

A.2.1 Observations during expert design session

During the expert study, we made the following observations:

- When designing the robot responses, we found that the height of the person greatly influences how a response is interpreted. For example, a small person giving a hug while the robot leans into the hug is perceived differently than a tall person giving a hug while the robot bends backwards. During the design phase and User Study 2, we will therefore ensure a consistent robot height for all participants (e.g., positioning the top of the robot between the chin and chest) by using a height-adjustable table.
- We assumed a neutral starting position from which the robot begins each response. This allows the robot to move easily in any direction and to return to its starting position in a smooth, continuous loop.
- The robot had less range of motion when fully extended, as the skirt restricted movement by being stretched. Conversely, a greater range of motion was achieved when the robot was smaller.
- In general, energetic responses were achieved by having the robot bounce up and down rapidly during (e.g., stroking) or after (e.g., a hug) an interaction. Calm responses were achieved using minimal motion changes and a slow return to the starting position, while defensive responses were achieved by having the robot turn away from the touch. For the exact responses, see the next chapter.
- The noise produced by the motor contributed to certain interactions by giving the robot a more lively and recognizable character. However, to ensure that participants focus solely on movement during the user study, the robot's noise should be blocked out.
- During the session, we found that only a few gestures benefited significantly from the inflation and air functionality. Most gestures and personalities remained clearly distinguishable without it.
- The calm personality was the hardest to design for, as it is difficult to create a response that carries no judgment or opinion and does not evoke emotion.
- We observed that the number of vertical bounces differentiated between expressions of happiness and anger. Anger was conveyed through larger, more forceful bounces sustained for slightly longer than comfortable.

A.3 Data and responses

The collected touch gestures from study 1 can be found in GitHub: <https://github.com/mdethmers/Social-Touch-Research>

The resulting TD behaviours can be found in the GitHub: <https://github.com/mdethmers/Social-Touch-Research>